

```
In [2]: import numpy as np
```

```
In [3]: ones_arr = np.ones((5,5), dtype=int)
```

```
In [4]: ones_arr
```

```
Out[4]: array([[1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1]])
```

```
In [5]: ones_arr * 255
```

```
Out[5]: array([[255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255]])
```

```
In [6]: import matplotlib.pyplot as plt
```

```
In [7]: from PIL import Image
```

```
In [8]: image = Image.open(r'C:\Users\bhara\Downloads\360_F_794113213_1n5N8pTx3NgQWhi3N6
```

```
In [9]: image
```

```
Out[9]:
```



```
In [10]: image1 = Image.open(r'C:\Users\bhara\Downloads\Pics\2134.jpg')
image1
```

Out[10]:

In [11]: `type(image1)`Out[11]: `PIL.JpegImagePlugin.JpegImageFile`In [12]: `image_arr = np.asarray(image)`
`image_arr`

```

Out[12]: array([[ 8, 12, 41],
                [ 9, 13, 42],
                [ 8, 12, 41],
                ...,
                [153, 62, 158],
                [155, 64, 160],
                [156, 65, 161]],

                [[ 8, 12, 41],
                [10, 14, 43],
                [ 9, 13, 42],
                ...,
                [151, 60, 156],
                [152, 61, 157],
                [154, 63, 159]],

                [[ 7, 11, 40],
                [11, 15, 44],
                [11, 15, 44],
                ...,
                [148, 57, 153],
                [149, 58, 154],
                [150, 59, 155]],

                ...,

                [[16,  6, 30],
                [16,  6, 30],
                [16,  6, 30],
                ...,
                [ 3,  8, 28],
                [ 4,  9, 29],
                [ 4,  9, 29]],

                [[15,  5, 29],
                [15,  5, 29],
                [16,  6, 30],
                ...,
                [ 5, 10, 30],
                [ 5, 10, 30],
                [ 5, 10, 30]],

                [[15,  5, 29],
                [15,  5, 29],
                [15,  5, 29],
                ...,
                [ 5, 10, 30],
                [ 5, 10, 30],
                [ 6, 11, 31]]], dtype=uint8)

```

```

In [13]: i2_arr = np.asarray(image1)
         i2_arr

```

```

Out[13]: array([[28, 30, 25],
               [26, 28, 23],
               [15, 20, 14],
               ...,
               [ 6, 15, 14],
               [ 7, 16, 15],
               [ 8, 17, 16]],

              [[ 6,  8,  3],
               [18, 20, 15],
               [27, 32, 26],
               ...,
               [ 3, 12, 11],
               [ 3, 12, 11],
               [ 3, 12, 11]],

              [[ 4,  6,  1],
               [19, 21, 16],
               [32, 37, 31],
               ...,
               [ 6, 14, 16],
               [ 7, 15, 17],
               [ 7, 15, 17]],

              ...,

              [[ 5, 15, 16],
               [ 8, 18, 19],
               [15, 25, 26],
               ...,
               [ 7, 19, 15],
               [ 7, 19, 15],
               [ 8, 20, 16]],

              [[ 7, 17, 18],
               [13, 23, 24],
               [19, 29, 30],
               ...,
               [ 8, 20, 16],
               [10, 22, 18],
               [12, 24, 20]],

              [[12, 22, 23],
               [13, 23, 24],
               [11, 21, 22],
               ...,
               [ 9, 21, 17],
               [10, 22, 18],
               [12, 24, 20]]], dtype=uint8)

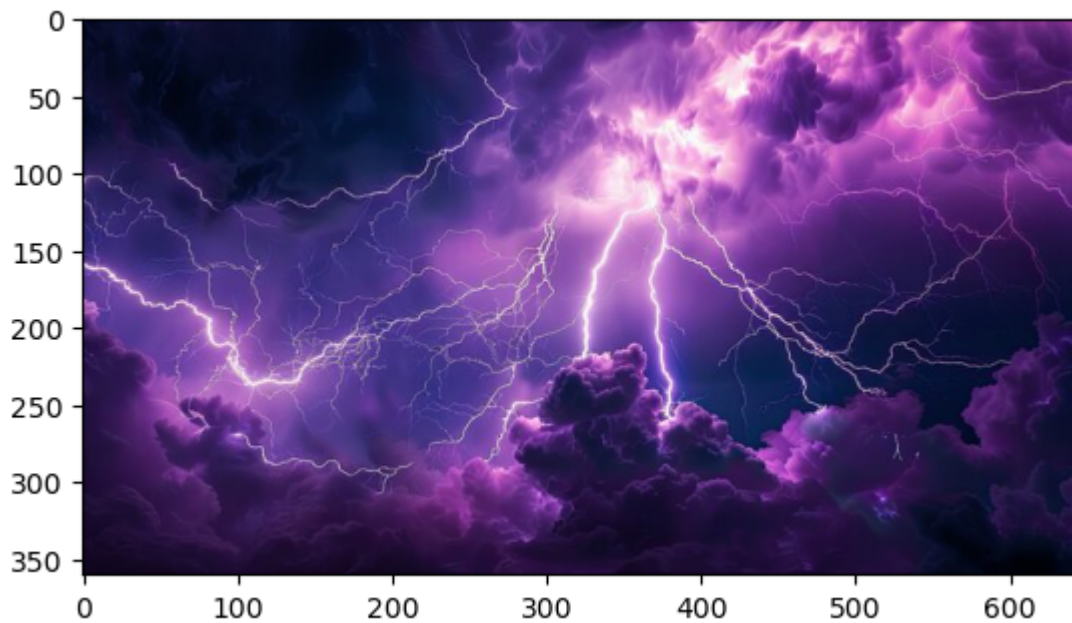
```

```
In [14]: type(i2_arr)
```

```
Out[14]: numpy.ndarray
```

```
In [15]: plt.imshow(image_arr)
```

```
Out[15]: <matplotlib.image.AxesImage at 0x1591e7b36b0>
```

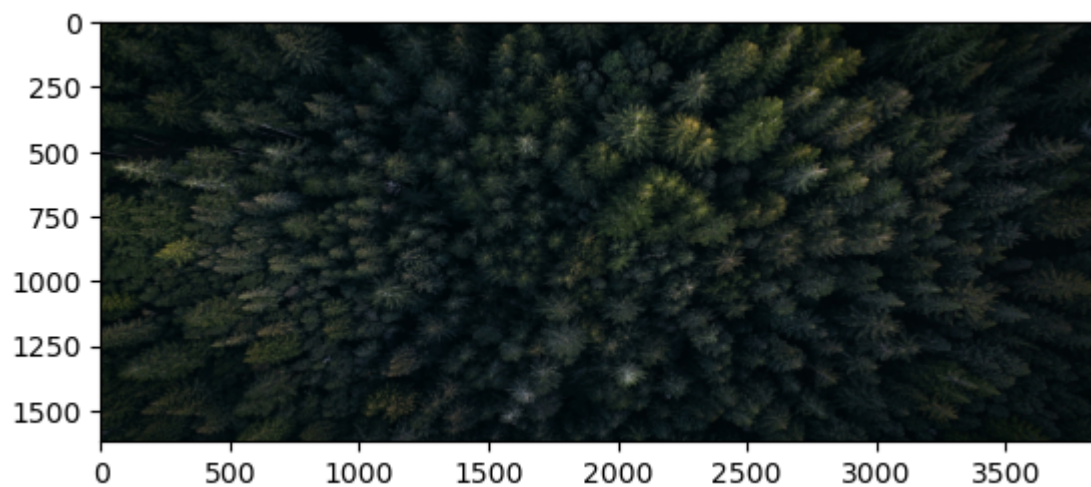


```
In [16]: image_arr.shape
```

```
Out[16]: (360, 643, 3)
```

```
In [17]: plt.imshow(i2_arr)
```

```
Out[17]: <matplotlib.image.AxesImage at 0x1591e841ac0>
```



```
In [18]: i2_arr.shape
```

```
Out[18]: (1620, 3840, 3)
```

```
In [19]: image_red=image_arr.copy()  
image_red
```

```

Out[19]: array([[ 8, 12, 41],
               [ 9, 13, 42],
               [ 8, 12, 41],
               ...,
               [153, 62, 158],
               [155, 64, 160],
               [156, 65, 161]],

            [[ 8, 12, 41],
             [10, 14, 43],
             [ 9, 13, 42],
             ...,
             [151, 60, 156],
             [152, 61, 157],
             [154, 63, 159]],

            [[ 7, 11, 40],
             [11, 15, 44],
             [11, 15, 44],
             ...,
             [148, 57, 153],
             [149, 58, 154],
             [150, 59, 155]],

            ...,

            [[16,  6, 30],
             [16,  6, 30],
             [16,  6, 30],
             ...,
             [ 3,  8, 28],
             [ 4,  9, 29],
             [ 4,  9, 29]],

            [[15,  5, 29],
             [15,  5, 29],
             [16,  6, 30],
             ...,
             [ 5, 10, 30],
             [ 5, 10, 30],
             [ 5, 10, 30]],

            [[15,  5, 29],
             [15,  5, 29],
             [15,  5, 29],
             ...,
             [ 5, 10, 30],
             [ 5, 10, 30],
             [ 6, 11, 31]]], dtype=uint8)

```

```
In [20]: image_arr == image_red
```

```

Out[20]: array([[ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True],
                ...,
                [ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True]],

               [[ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True],
                ...,
                [ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True]],

               [[ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True],
                ...,
                [ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True]],

               ...,

               [[ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True],
                ...,
                [ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True]],

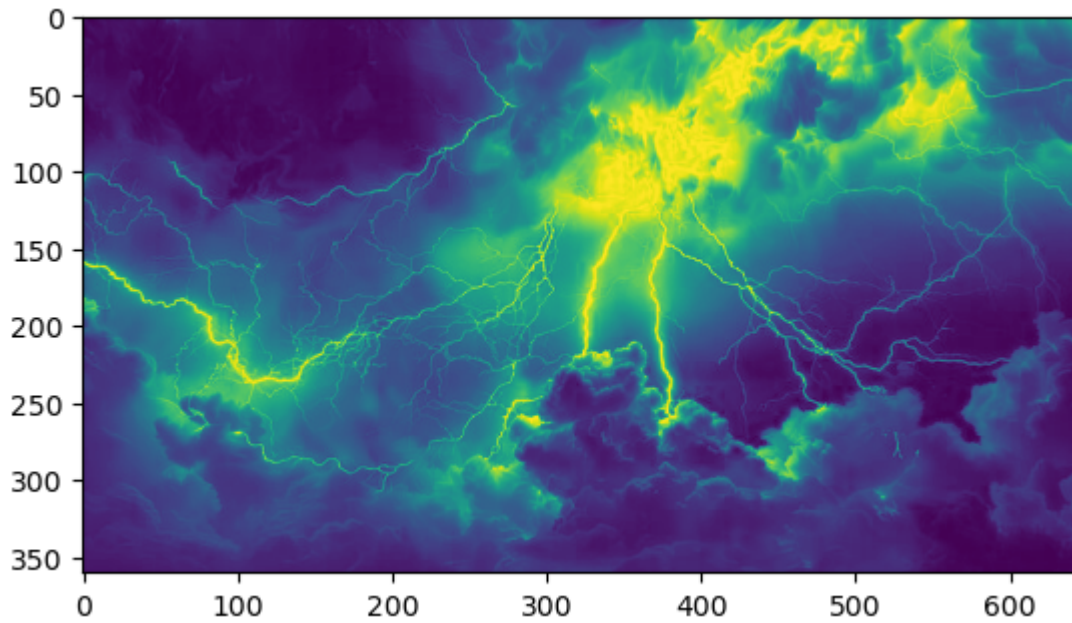
               [[ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True],
                ...,
                [ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True]],

               [[ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True],
                ...,
                [ True,  True,  True],
                [ True,  True,  True],
                [ True,  True,  True]]])

```

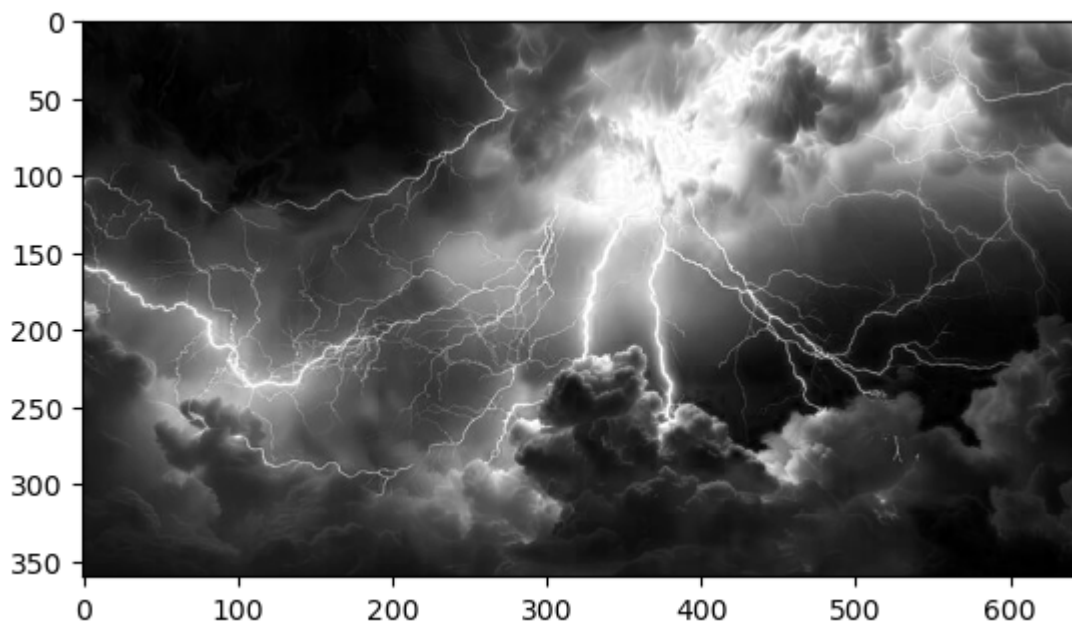
```
In [21]: plt.imshow(image_red[:, :, 0])
```

```
Out[21]: <matplotlib.image.AxesImage at 0x1591e810dd0>
```



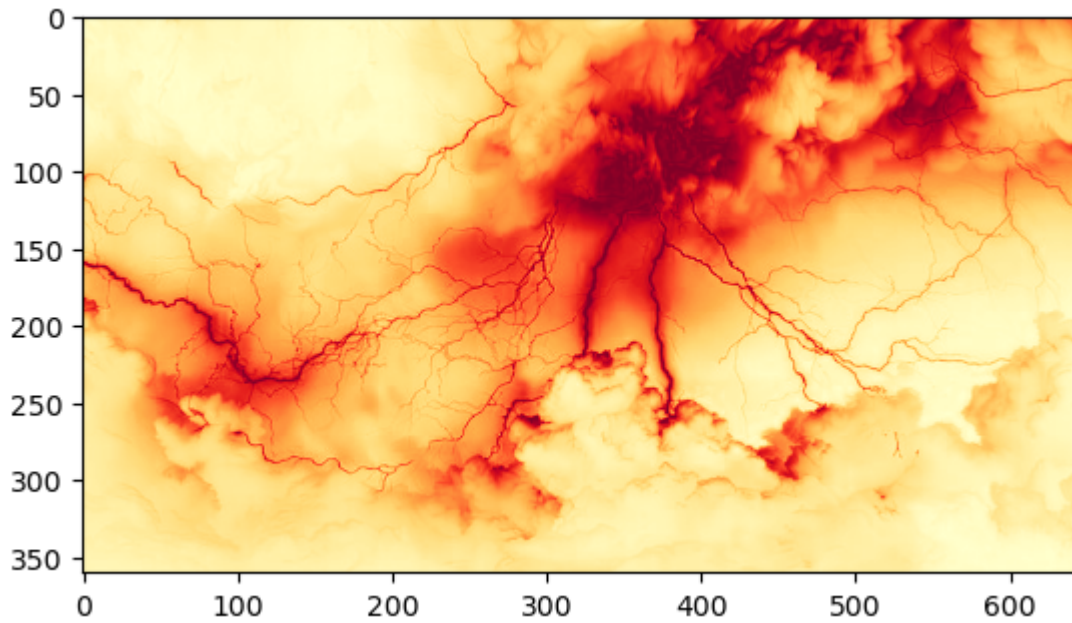
```
In [22]: plt.imshow(image_red[:, :, 0], cmap='gray')
```

```
Out[22]: <matplotlib.image.AxesImage at 0x1591e86a960>
```



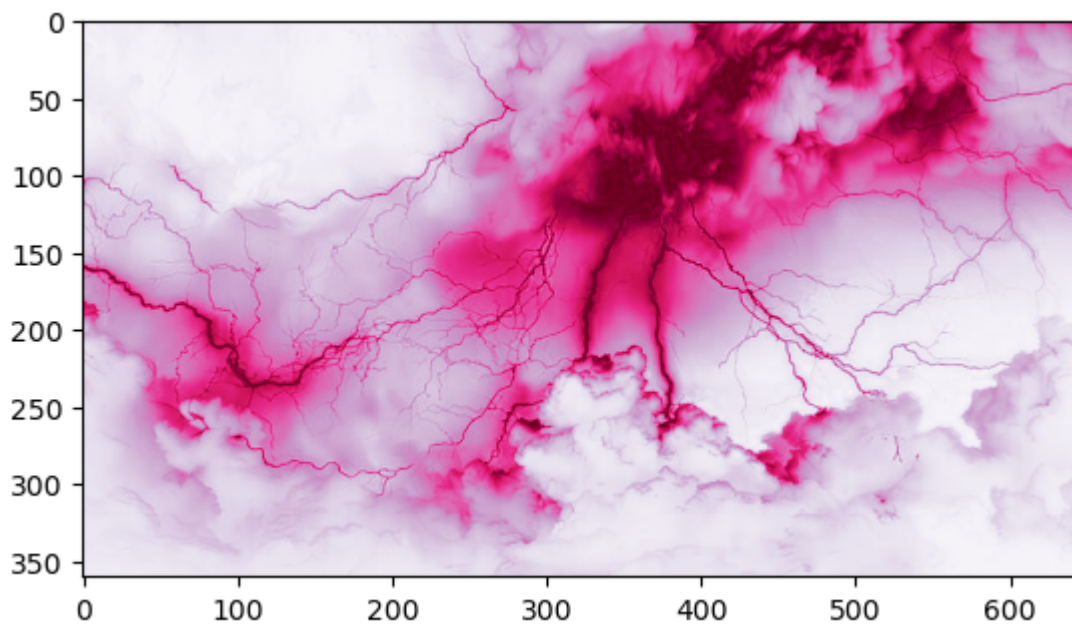
```
In [23]: plt.imshow(image_red[:, :, 0], cmap='YlOrRd')
```

```
Out[23]: <matplotlib.image.AxesImage at 0x1592476c5f0>
```



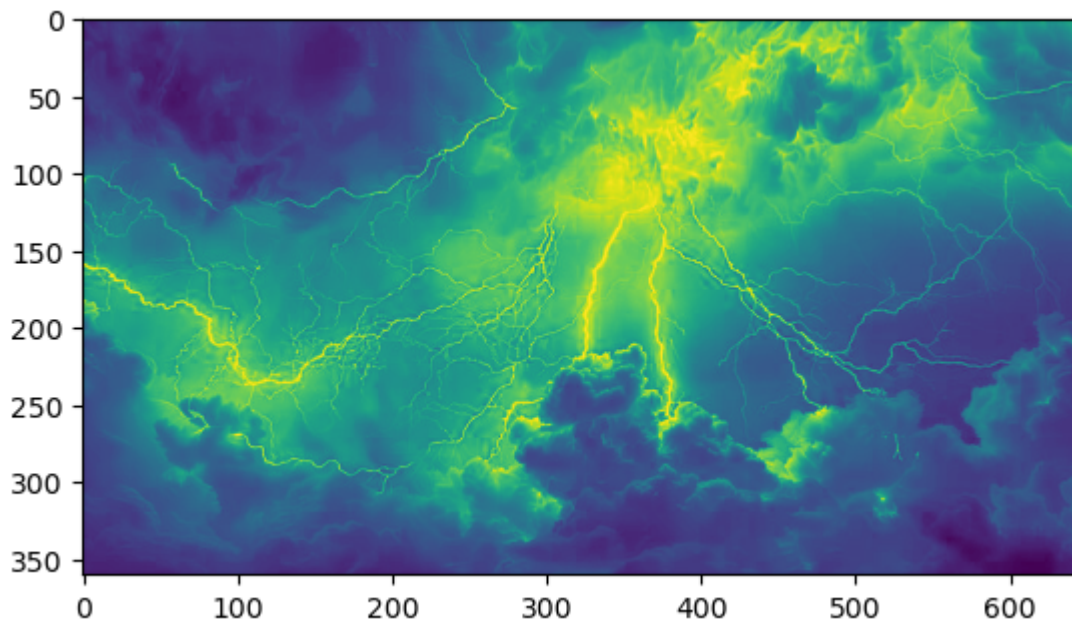
```
In [24]: plt.imshow(image_red[:, :, 0], cmap = 'PuRd')
```

```
Out[24]: <matplotlib.image.AxesImage at 0x159247ac140>
```



```
In [25]: plt.imshow(image_red[:, :, 2])
```

```
Out[25]: <matplotlib.image.AxesImage at 0x1591fae0110>
```



```
In [26]: image_red[:, :, 1]
```

```
Out[26]: array([[12, 13, 12, ..., 62, 64, 65],  
               [12, 14, 13, ..., 60, 61, 63],  
               [11, 15, 15, ..., 57, 58, 59],  
               ...,  
               [ 6,  6,  6, ...,  8,  9,  9],  
               [ 5,  5,  6, ..., 10, 10, 10],  
               [ 5,  5,  5, ..., 10, 10, 11]], dtype=uint8)
```

```
In [27]: plt.imshow(image_red, cmap='Accent')
```

```
Out[27]: <matplotlib.image.AxesImage at 0x1592475aed0>
```

